

# Research Lab Management System (RLMS)

## Lab 1: Conceptual Design Report

**Course:** CS M211 - Data Management System — **Instructor:** Prof. Karima Echihabi

**Institution:** UM6P - College of Computing — **Date:** September 20, 2026

### 1. Team Roles and Contributions

| Member Name | Role                                    | Primary Responsibilities & Deliverables                                                                                                                                                                  |
|-------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Abdelhalim  | Technical Designer & Diagram Specialist | Responsible for drawing the complete ER diagram in Draw.io. Managed software tools, layout, structural icons, and visual updates.                                                                        |
| Hamza       | Co-Requirements Analyst & Modeler       | Helped design the model and analyze what we needed. Worked on defining the main items, connecting them, creating subcategories, and deciding what details belong to each.                                |
| Reda        | Co-Requirements Analyst & Modeler       | Helped design the model and analyze what we needed. Worked on figuring out connection rules (like one-to-many), identifying dependent items, grouping things together, and setting rules for unique IDs. |

### 2. Key Challenges and Resolution Strategies

- **Grouping Funded Items vs. Direct Links (BD3):**  
*Challenge:* How to show budgets that fund both projects and labs without making the diagram messy or redundant.  
*Resolution:* We grouped the funded items together. This allowed us to use a single “Funded” connection with the Budget while keeping our connection rules accurate.
- **Handling Complex Reservations Involving Multiple Parts (RE1):**  
*Challenge:* Deciding whether a reservation should be one massive connection or broken down into simpler, separate links.  
*Resolution:* We broke it down into simpler connections linking the Reservation to the Person, Research Project, and Equipment Unit. This keeps the diagram clear while showing exactly who made the reservation and what it's for.
- **Supplier Rules vs. Daily Usage (CO4):**  
*Challenge:* Separating the special supplier prices negotiated for each lab from the everyday use of lab supplies.  
*Resolution:* We created a three-way connection (Supplies) between the Supplier, Consumable, and Laboratory to track the specific unit price, keeping it structurally separate from everyday reservations.

### 3. Modeling Choices and Justifications

- **Two-Level Categories (PE2, PE4):** We set up Person as a main category that splits into Student and Employee. Employee then splits into Academic and Non-Academic staff. This distinction helps us easily apply rules, like only allowing academic staff to manage budgets or supervise projects.
- **Grouping for Funding and Stock Monitoring (BD3, CO8):** We grouped projects and laboratories together so the budget can fund them clearly. We also grouped the Laboratory and Consumable connection (Stock) so that Technical Staff can monitor the stock directly via the Monitored connection.
- **Dependent Items for Calibration Records (CA1–CA4):** We made Calibration Record a dependent item because it cannot exist without its main Equipment Unit. It is uniquely identified by its date combined with its equipment's ID.
- **Three-Way Connection for Supplies (CO4):** We used a three-way connection linking Supplier, Consumable, and Laboratory. This lets us record the specific price for each lab-supplier pair, as pricing varies depending on the lab.

### 4. Assumptions

- **Supply Reservation Context (CO10):** Assumed that a supply item is linked to **at most one** active reservation at a time, which is why we used a direct link to Reservation.
- **Supervisor Dynamics (PE5):** Assumed that each employee has **at most one** immediate boss (supervisor), while a boss can manage many employees.
- **Equipment Location Updates (EQ4):** Assumed that an equipment unit belongs to **exactly one** laboratory at any given time. When it moves, we just update the location instead of keeping a history of its past locations.

## 5. Limitations of the Diagram

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Standard ER diagrams cannot show dynamic business rules. These rules must be handled later using database code or application software:

- **Single Role per Project (PR3, PR4):** The diagram cannot force a person to strictly have only one role per research project.
- **Checking Certifications (CE5, CE7):** The diagram cannot automatically verify that a person using an equipment unit actually holds the required safety certification for that specific model.
- **Reservation Time Overlaps (RE4):** We cannot visually show rules that prevent two people from reserving the same equipment at the same exact time.
- **Active Reservation for Supplies (CO5):** We cannot use structural lines to enforce the rule that supplies can only be requested if there is an active and valid equipment reservation.